INCM PROPOSAL Topic 27: Evidence Accumulation

# Modeling Simple Decisions & Speed-Accuracy Trade-off

A 1D Drift Diffusion Model analysis across task difficulties and dynamic thresholds

### Evidence Accumulation

Continuous noisy integration until reaching decision thresholds.

### Speed-Accuracy Trade-off

High caution improves accuracy but increases response latencies.

### Collapsing Boundaries

Dynamic urgency signal prevents long RT tails on hard choices.

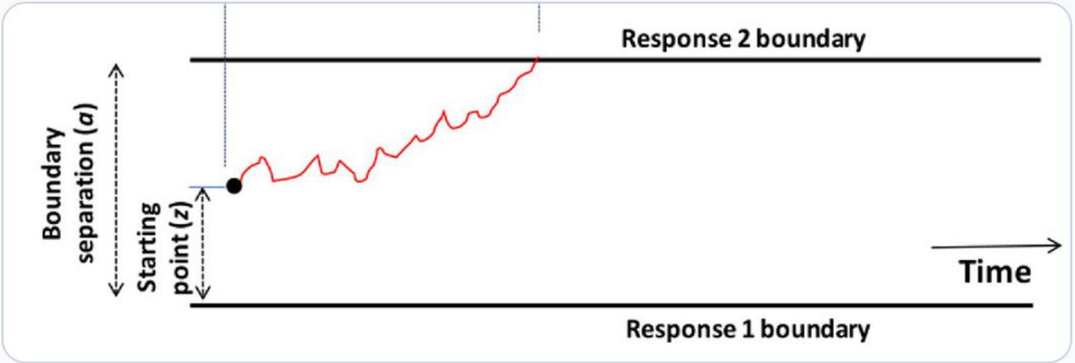
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CORE RESEARCH OBJECTIVES

# Problem Statement & Key Questions

### ⚠ Flat Boundary Bottleneck

Standard fixed-boundary DDMs predict unrealistically long decision latencies on zero-contrast trials. Biological decision-makers employ dynamic urgency to force decisions under severe uncertainty.




QUESTION 1

### Curve Reproduction

Standard DDM fit across task difficulty levels.

QUESTION 2

### Boundary Dynamics

Collapsing threshold impact on zero-contrast trials.

QUESTION 3

### Biological Validity

Evaluating motor delay and noise assumptions.

QUANTITATIVE ARCHITECTURE

# Methodology & Simulation Protocol

### 1D Diffusion Stochastic Process

```
x(t+Δt) = x(t) + v(c)Δt + σ√Δt ξ(t)
v(c) = k · c_i | ξ(t) ~ N(0, 1)
RT = T_d + t_0
```

**Collapsing Boundary Extension**

$\theta(t) = \pm a \cdot \exp(-(t / \lambda)^{\gamma})$

**Euler–Maruyama Integration**
1ms step size with 3–second maximum duration cutoff.

**Parameter Calibration**
Nelder-Mead simplex fitting on choice fraction and RT distributions.

**Model Evaluation Metrics**
Psychometric, chronometric, and BIC goodness-of-fit comparison.

| 1 ms | 6 Levels | BIC / Likelihood |
|---|---|---|
| Simulation Step (Δt) | Visual Contrast Range | Boundary Selection Metric |

FOUNDATIONS & BENCHMARKS

# Data Repositories & Key References

## Data Repositories

**IBL Mouse Visual 2AFC Dataset**
International Brain Laboratory (2021) public behavioral repository.

- **Contrast Levels:** 0%, 6.25%, 12.5%, 25%, 50%, 100%
- **Trial Filtering:** Exclude RT < 100ms and > 2.5s
- **Synthetic Data:** Known parameter recovery verification

## Key Literature

**Ratcliff & McKoon (2008)** — DDM theory and two-choice decision dynamics.

**Bogacz et al. (2006)** — Mathematical SPRT equivalence and optimality.

**Drugowitsch et al. (2012)** — Evidence accumulation costs and urgency signals.

**IBL (2021)** — Standardized mouse decision-making dataset in eLife.

Implementation Framework: Python 3 with NumPy, SciPy (Nelder-Mead), and Matplotlib visualization pipelines.

EXECUTION PLAN

# Project Timeline & Milestones

WEEK 1

### Simulator Development

Implement Python Euler-Maruyama simulator and test synthetic parameter recovery.

WEEK 2

### Data Pipeline & Fitting

Process IBL mouse dataset and fit fixed-boundary DDM via Nelder-Mead optimization.

WEEK 3

### Collapsing Boundary

Implement exponential threshold decay and perform BIC model comparison.

WEEK 4

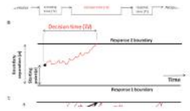
### Evaluation & Report

Generate psychometric & chronometric curves and finalize project deliverable.

Primary Deliverable: Finalized Python Modeling Suite & Course Presentation

INCM 2026

# Image Sources



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